Spatial Linear Models

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In this illustration, we will learn how to fit spatial linear models by using phosphorus example B. See Figure 1.

Inorganic P (ppm) *(explanatory variable)*			
28.0	6.0	8.6	11.4
17.5	14.0	12.4	27.3
16.8	9.9	30.1	29.9

Organic P (ppm) *(response variable)*			
37.9	8.7	16.5	22.1
22.6	20.6	22.6	46.0
20.4	14.1	46.5	44.2

Figure 1: Phosphorus B

Creating Neighbors

We first create the data vectors for inorganic phosphorous, organic phosphorous, and plot index.

```
library(spdep)
library(spatialreg)
```

```
# Example B
phos.inorganic2 <- c(28.0, 6.0, 8.6, 11.4,
                     17.5, 14.0, 12.4, 27.3,
                     16.8, 9.9, 30.1, 29.9)
# explanatory variable

phos.organic2 <- c(37.9, 8.7, 16.5, 22.1,
                   22.6, 20.6, 22.6, 46.0,
                   20.4, 14.1, 46.5, 44.2)
# response

plot2 <- as.factor(seq(1, 12))  # plot index
```

We check the data vectors for inorganic and organic phosphorous against Figure 2 by the function `matrix()`.

```
matrix(phos.inorganic2, ncol = 4, byrow = TRUE)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 28.0  6.0  8.6 11.4
## [2,] 17.5 14.0 12.4 27.3
## [3,] 16.8  9.9 30.1 29.9
```

```
matrix(phos.organic2, ncol = 4, byrow = TRUE)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 37.9  8.7 16.5 22.1
```

```
## [2,] 22.6 20.6 22.6 46.0
## [3,] 20.4 14.1 46.5 44.2
```

Having verified that the data were entered correctly, we create the data frame `phosB` using the `cbind.data.frame()` function to column bind the data vectors `phos.inorganic2`, `phos.organic2`, and `plot2` together.

```
phosB <- cbind.data.frame(phos.inorganic2, phos.organic2, plot2)
```

We generate the first-order neighborhood structure object using the `cell2nb()` function from the `spdep` package and check the structure.

```
# Generate neighbourhood structure object for grid plots
phos.nb <- cell2nb(3, 4, type = "rook")
str(phos.nb)
```

```
## List of 12
##  $ : int [1:2] 2 5
##  $ : int [1:3] 1 3 6
##  $ : int [1:3] 2 4 7
##  $ : int [1:2] 3 8
##  $ : int [1:3] 1 6 9
##  $ : int [1:4] 2 5 7 10
##  $ : int [1:4] 3 6 8 11
##  $ : int [1:3] 4 7 12
##  $ : int [1:2] 5 10
##  $ : int [1:3] 6 9 11
##  $ : int [1:3] 7 10 12
##  $ : int [1:2] 8 11
##  - attr(*, "class")= chr "nb"
##  - attr(*, "call")= language cell2nb(nrow = 3, ncol = 4, type = "rook")
##  - attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##  - attr(*, "cell")= logi TRUE
##  - attr(*, "rook")= logi TRUE
##  - attr(*, "sym")= logi TRUE
```

We also generate the list of spatial weights using `nb2listw()` from the `spdep` package. We specify `style = "B"` for binary weights and check the structure using `str()`.

```
# Create list of spatial weights for neighbourhood structure object using binary weights
phos.lw2 <- nb2listw(phos.nb, style = "B")
str(phos.lw2)
```

```
## List of 3
##  $ style     : chr "B"
##  $ neighbours:List of 12
##   ..$ : int [1:2] 2 5
##   ..$ : int [1:3] 1 3 6
##   ..$ : int [1:3] 2 4 7
##   ..$ : int [1:2] 3 8
##   ..$ : int [1:3] 1 6 9
##   ..$ : int [1:4] 2 5 7 10
##   ..$ : int [1:4] 3 6 8 11
##   ..$ : int [1:3] 4 7 12
##   ..$ : int [1:2] 5 10
##   ..$ : int [1:3] 6 9 11
##   ..$ : int [1:3] 7 10 12
```

```
##   ..$ : int [1:2] 8 11
##   ..- attr(*, "class")= chr "nb"
##   ..- attr(*, "call")= language cell2nb(nrow = 3, ncol = 4, type = "rook")
##   ..- attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##   ..- attr(*, "cell")= logi TRUE
##   ..- attr(*, "rook")= logi TRUE
##   ..- attr(*, "sym")= logi TRUE
##  $ weights   :List of 12
##   ..$ : num [1:2] 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:2] 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:4] 1 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:2] 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:3] 1 1 1
##   ..$ : num [1:2] 1 1
##   ..- attr(*, "mode")= chr "binary"
##   ..- attr(*, "B")= logi TRUE
##  - attr(*, "class")= chr [1:2] "listw" "nb"
##  - attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##  - attr(*, "call")= language nb2listw(neighbours = phos.nb, style = "B")
##  - attr(*, "zero.policy")= logi FALSE
```

Spatial Linear Model

Now that we have created the neighborhood structure object and list of spatial weights, we apply spatial linear regression. We will be using the function `spautolm()` from the `spatialreg` package. We will fit two models for the error term in the spatial linear regression model: the spatial autoregressive or SAR error and the conditional autoregressive or CAR error.

SAR

Recall that the spatial linear regression model with SAR error is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where

$$\varepsilon_i = \rho \sum_{j=1}^{n} s_{ij}\varepsilon_j + u_i.$$

Here, u_i are independent and follow a normal distribution with mean zero and variance $\sigma^2 v_i$. The quantity s_{ij} takes on a 1 where i and j are neighbors for all $i \neq j$, and a zero otherwise.

$$s_{ij} = \begin{cases} 1; & i \neq j, \text{ and } i \text{ and } j \text{ are neighbors,} \\ 0; & i = j, \text{ or } i \text{ and } j \text{ are not neighbors.} \end{cases}$$

$$[u_i \sim \text{iid} N(0, \sigma^2 v_i)]$$

In order to fit this model, we will use the function `spautolm()`, which has similar syntax to the function `lm()` used for linear regression. However, we need to specify the type of error term using `family = "SAR"` and the list of spatial weights by setting `listw = phos.lw2`. Below, we fit the spatial linear model with SAR error where `phos.organic2` is the response and `phos.inorganic2` is the explanatory variable.

```
# Fit the spatial linear regression with SAR error
phos.slm <- spautolm(phos.organic2 ~ phos.inorganic2,
                     family = "SAR",
                     listw = phos.lw2,
                     data = phosB)
```

We can check the model fit output using the `summary()` function.

```
summary(phos.slm)
```

```
##
## Call: spautolm(formula = phos.organic2 ~ phos.inorganic2, data = phosB,
##     listw = phos.lw2, family = "SAR")
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -2.294762 -0.786550 -0.083627  1.217807  3.259397
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -1.91246    2.44295 -0.7828   0.4337
## phos.inorganic2  1.45088    0.05958 24.3518   <2e-16
##
## Lambda: 0.30159 LR test value: 11.202 p-value: 0.00081704
## Numerical Hessian standard error of lambda: 0.026746
##
## Log likelihood: -25.75411
## ML residual variance (sigma squared): 2.7528, (sigma: 1.6592)
## Number of observations: 12
## Number of parameters estimated: 4
## AIC: 59.508
```

From the first row of the summary output, we have summary statistics on the model residuals, where the minimum is -2.294762, the first quartile is -0.786550, the median is -0.083627, the third quartile is 1.217807, and the maximum is 3.259397.

Next, we have the regression coefficient estimates, standard errors, z-values, and p-values, the same as we have for `lm` output. Here, the fitted intercept is -1.91246, and the fitted slope for inorganic phosphorous is 1.45088; the p-value for inorganic phosphorous is less than 2×10^{-16}.

In the following line, we have that the fitted lambda value is 0.30159. A model with no spatial correlation would have lambda equal to 0. The likelihood ratio (or LR) test value is 11.202. The likelihood ratio test here compares the model with no spatial correlation, where lambda is equal to 0, to the model which allows for spatial correlation, which is the fitted model with the non-zero lambda term. The p-value for this test is 0.00081704; there is strong evidence of spatial correlation in the error term. The output also provides the residual variance or overall model error as 2.7528. Lastly, the summary output provides the number of observations (12), the number of parameters estimated (4), and the AIC value (59.508).

Using Moran's I, we could test for residual spatial correlation using `moran.test()` on the model residuals.

```
set.seed(7)
moran.test(resid(phos.slm), phos.lw2, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(phos.slm)
## weights: phos.lw2
##
## Moran I statistic standard deviate = 1.4989, p-value = 0.1339
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.22006830       -0.09090909        0.04304448
```

We see the Moran's I statistic is 0.22006830 and the p-value is 0.1339. At the 10% alpha level, we do not reject the null hypothesis of CSR. There is no evidence of remaining spatial correlation. The SAR error model is adequate to account for the spatial dependence in the data.

We can then interpret the linear model as we normally would: an increase of inorganic phosphorous by one part per million is associated with an increase of 1.45088 ppm in the organic phosphorous.

CAR

The spatial linear regression model with CAR error is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where the ε_i follows a normal distribution.

The expectation of ε_i conditional on ε_j where $j \neq i$ equals ρ times the summation from $j = 1$ to n of c_{ij} times ε_j. The variance of ε_i conditional on ε_j where $j \neq i$ equals $\sigma^2 v_i$.

$$E(\varepsilon_i \mid \varepsilon_j : j \neq i) = \rho \sum_{j=1}^{n} c_{ij} \varepsilon_j,$$

$$\text{Var}(\varepsilon_i \mid \varepsilon_j : j \neq i) = \sigma^2 v_i,$$

where

$$c_{ij} = \begin{cases} 1; & i \neq j, \text{ and } i \text{ and } j \text{ are neighbors,} \\ 0; & i = j, \text{ or } i \text{ and } j \text{ are not neighbors.} \end{cases}$$

Additionally, we require that

$$c_{ij} v_j = c_{ji} v_i.$$

In order to fit this model, we will use the function `spautolm()` and specify `family = "CAR"` and the list of spatial weights by setting `listw = phos.lw2`. Below, we fit the model where `phos.organic2` is the response and `phos.inorganic2` is the explanatory variable.

```
# Fit the spatial linear regression with CAR error
phos.clm <- spautolm(phos.organic2 ~ phos.inorganic2,
                     family = "CAR",
                     listw = phos.lw2,
                     data = phosB)
```

We can check the model output using the `summary()` function.

```
summary(phos.clm)
```

```
##
## Call: spautolm(formula = phos.organic2 ~ phos.inorganic2, data = phosB,
##     listw = phos.lw2, family = "CAR")
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.76169 -0.92436 -0.20615  0.96683  3.08927
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -0.349645   2.735238 -0.1278   0.8983
## phos.inorganic2  1.436211   0.081622 17.5958   <2e-16
##
## Lambda: 0.31595 LR test value: 6.7189 p-value: 0.0095395
## Numerical Hessian standard error of lambda: 0.025852
##
## Log likelihood: -27.99571
## ML residual variance (sigma squared): 4.6705, (sigma: 2.1611)
## Number of observations: 12
## Number of parameters estimated: 4
## AIC: 63.991
```

In the output above, the first row of the summary output shows the summary statistics of the residuals provided are the minimum (-2.76169), the first quartile (-0.92436), the median (-0.20615), the third quartile (0.96683), and the maximum (3.08927). The fitted intercept term is -0.349645, and the fitted slope for inorganic phosphorous is 1.436211; the p-value for inorganic phosphorous is less than 2×10^{-16}.

In the following line, we have the fitted lambda value is 0.31595. The likelihood ratio (or LR) test value is 6.7189. The likelihood ratio test compares the model with no spatial correlation, where lambda is equal to 0, to the model which allows for spatial correlation with the non-zero lambda term. The p-value is 0.0095395; there is strong evidence of spatial correlation in the error term. The residual variance or overall model error is 4.6705. The number of observations is 12, the number of parameters estimated is 4, and the AIC value is 63.991.

Using Moran's I, we could test for residual spatial correlation using `moran.test()` on the model residuals.

```
set.seed(8)
moran.test(resid(phos.clm), phos.lw2, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(phos.clm)
## weights: phos.lw2
##
## Moran I statistic standard deviate = 1.0475, p-value = 0.2949
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.12676367       -0.09090909        0.04318399
```

The observed Moran's I statistic is 0.12676367 and the p-value is 0.2949. At the 10% alpha level, we do not reject the null hypothesis of CSR. We find no evidence of residual spatial correlation. The CAR error model is adequate to account for the spatial dependence in the data.

We can interpret the linear model as we normally would: an increase of inorganic phosphorous by one ppm is associated with an increase by 1.436211 ppm in the organic phosphorous.